Pseudo Classes:

A CSS ***pseudo-class*** is a keyword added to a selector that lets you style a specific state of the selected element.

Characteristics of Pseudo Classes:

Syntax: Pseudo-classes are declared using a single colon (:) followed by the name of the pseudo-class, appended to a selector.

```
selector:pseudo-class {
  property: value;
}
```

Purpose: They target elements only when they are in a specific state, such as when they are being hovered over, when they have been visited, or when they are the first child of their parent.

1. User Action (Dynamic) Pseudo-Classes

These respond to user interaction.

Pseudo-Class	Description	Example
`:hover`	When user places the mouse pointer over an element	`a:hover { color: red; }`
`:active`	When an element is being clicked	`button:active { transform: scale(0.9); }`
`:focus`	When an element (like input) is focused	`input:focus { border-color: blue; }`
`:visited`	Applies to links that have been visited	`a:visited { color: purple; }`
`:link`	Applies to links that haven't been visited	`a:link { color: blue; }`

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>User Action Pseudo-Classes</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            padding: 40px;
            background-color: #f4f4f4;
        }

        h2 {
            color: #333;
        }
        a:link {
            color: blue;
            text-decoration: none;
        }

        a:visited {
            color: purple;
        }

        a:hover {
            color: white;
            background-color: blue;
            padding: 5px 10px;
            border-radius: 4px;
        }

        a:active {
            background-color: darkblue;
        }

        input[type="text"] {
            padding: 8px;
            border: 2px solid #ccc;
            border-radius: 4px;
            outline: none;
            transition: border-color 0.3s, box-shadow 0.3s;
        }
```

```
        input[type="text"]:focus {
            border-color: dodgerblue;
            box-shadow: 0 0 5px dodgerblue;
        }

        button {
            padding: 10px 20px;
            background-color: #28a745;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
            transition: transform 0.2s, background-color 0.3s;
        }

        button:hover {
            background-color: #218838;
        }

        button:active {
            transform: scale(0.95);
            background-color: #1e7e34;
        }
    </style>
</head>

<body>

    <h2>User Action Pseudo-Classes </h2>
    <p>
        <a href="https://www.google.com" target="_blank">Visit Google</a>
    </p>
    <p>
        <label>Enter your name:</label><br>
        <input type="text" placeholder="Click or focus here">
    </p>
    <p>
        <button>Click Me</button>
    </p>
</body>
</html>
```

2. Structural Pseudo-Classes

These select elements based on their position in the DOM tree.

Pseudo-Class	Description	Example
`:first-child`	Selects the first child of a parent	`p:first-child { color: red; }`
`:last-child`	Selects the last child of a parent	`p:last-child { color: blue; }`
`:nth-child(n)`	Selects elements based on position (1-based index)	`li:nth-child(2) { color: green; }`
`:nth-last-child(n)`	Same as above, but counting from the end	`li:nth-last-child(1)`
`:only-child`	Selects elements that are the only child of their parent	`p:only-child { font-style: italic; }`
`:first-of-type`	Selects the first element of a specific type	`p:first-of-type { color: orange; }`
`:last-of-type`	Selects the last element of a specific type	`p:last-of-type { color: pink; }`
`:nth-of-type(n)`	Selects nth element of a given type	`p:nth-of-type(2)`
`:empty`	Selects elements with no children	`div:empty { display: none; }`

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Structural Pseudo-Class Example</title>
```

```
<style>
  ul {
    list-style: none;
    padding: 0;
    width: 200px;
    font-family: Arial, sans-serif;
  }

  ul li {
    margin: 5px 0;
    padding: 10px;
    border-radius: 4px;
    text-align: center;
  }

  ul li:nth-child(odd) {
    background-color: #bbdefb;
  }

  ul li:nth-child(even) {
    background-color: #e3f2fd;
  }

  ul li:first-child {
    background-color: #b9f6ca;
    font-weight: bold;
  }

  ul li:last-child {
    background-color: #ff8a80;
    font-weight: bold;
  }
</style>
</head>
<body>

<h3>Structural Pseudo-Classes with List</h3>

<ul>
  <li>Item 1 (first child)</li>
  <li>Item 2</li>
  <li>Item 3</li>
  <li>Item 4</li>
  <li>Item 5 (last child)</li>
</ul>
```

```
</body>
</html>
```

3. Form Pseudo-Classes

These target elements based on their state in a form or other user interface context.

Pseudo-Class	Description	Example
:checked	Selects radio buttons or checkboxes that are checked.	input:checked + label { color: green; }
:enabled	Selects user interface elements (like <input>, <button>) that are enabled.	input:enabled { opacity: 1; }
:disabled	Selects user interface elements that are disabled.	input:disabled { opacity: 0.5; }
:required	Selects form elements that have the required attribute.	input:required { border-left: 5px solid red; }

:valid, :invalid	Selects form elements whose content is valid or invalid based on the input type (e.g., email format).	input:invalid { background: pink; }

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Form Pseudo-Class Example</title>
<style>
  body {
    font-family: Arial, sans-serif;
    background: #f5f5f5;
    padding: 30px;
  }

  form {
    width: 300px;
    margin: auto;
    background: white;
    padding: 20px;
    border-radius: 8px;
  }

  label {
    display: block;
    margin-top: 10px;
  }

  input {
    width: 100%;
    padding: 8px;
    margin-top: 4px;
    border: 2px solid #ccc;
    border-radius: 4px;
    outline: none;
  }

  input:focus {
    border-color: dodgerblue;
```

```
  }

  input:required {
    border-left: 4px solid red;
  }

  input:valid {
    border-color: green;
  }

  input:invalid {
    border-color: red;
  }

  input[type="checkbox"]:checked + label {
    color: green;
    font-weight: bold;
  }

  button {
    margin-top: 15px;
    width: 100%;
    padding: 8px;
    background: #007bff;
    color: white;
    border: none;
    border-radius: 4px;
  }
</style>
</head>
<body>

<form>
  <label for="name">Name (required):</label>
  <input type="text" id="name" name="name" required>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" placeholder="example@mail.com">

  <input type="checkbox" id="agree">
  <label for="agree">I agree</label>

  <button type="submit">Submit</button>
</form>
```

```
</body>
</html>
```

4. Negation and Logical Pseudo-Classes

Pseudo-Class	Description	Example
`:not(selector)`	Excludes elements matching the selector	`p:not(.highlight) { color: gray; }`
`:is(selector)`	Matches if element fits *any* of listed selectors	`:is(h1, h2, h3) { font-weight: bold; }`
`:where(selector)`	Similar to `:is()`, but has no specificity	`:where(section, article) { margin: 20px; }`
`:has(selector)`	Selects element containing another matching element	`div:has(img) { border: 1px solid #333; }`

```
<!DOCTYPE html>
<html lang="en">
<head>
<style>
  div {
    width: 200px;
    padding: 10px;
    margin: 8px;
    text-align: center;
    border: 2px solid #999;
  }

  div:not(.special) {
    background-color: red;
  }

</style>
</head>
```

```
<body>

<h2>:not() Pseudo-Class Example</h2>

<div>Box 1</div>
<div class="special">Box 2 (special)</div>
<div>Box 3</div>

</body>
</html>
```

Tooltips:

A `tooltip` is a contextual text bubble that displays a description for an element that appears on pointer hover or keyboard focus

```
<!DOCTYPE html>
<html>
<style>
    .tooltip {
        position: relative;
        display: inline-block;
        border-bottom: 1px dotted black;
        cursor: pointer;
    }

    .tooltiptext {
        visibility: hidden;
        width: 130px;
        background-color: black;
        color: #ffffff;
        text-align: center;
        border-radius: 6px;
        padding: 5px 0;
        position: absolute;
        z-index: 1;
        bottom: 100%;
        left: 65%;
        margin-left: -65px;
    }

    .tooltip:hover .tooltiptext {
        visibility: visible;
```

```
    }
</style>

<body>

    <h2>Top-aligned Tooltip</h2>
    <p>Move the mouse over the text below:</p>

    <div class="tooltip">Hover over me
        <span class="tooltiptext">Some tooltip text</span>
    </div>

</body>

</html>
```

Images:

In **CSS**, an **image** is any **graphic resource** (like a photo, icon, or pattern) that is used as a **style** — not as page content.

It is not placed in the HTML `<img>` tag but is **applied through CSS properties** to decorate or enhance elements.

```
<!DOCTYPE html>
<html>

<head>
    <style>
        img {
            display: block;
            margin: auto;
            width: 20%;
        }
    </style>
</head>

<body>
    <h2>Center an Image Horizontally</h2>
```

```
    <p>To center an image horizontally, set margin to auto, convert it into a
block element, and define a width:</p>
    <img
src="https://fastly.picsum.photos/id/237/200/300.jpg?hmac=TmmQSbShHz9CdQm0NkEjx
1Dyh_Y984R9LpNrpvH2D_U" alt="Paris">
    <img src="" alt="" srcset="">
</body>

</html>
```

Image Filter :

The `filter` CSS property applies graphical effects like blur or color shift to an element. Filters are commonly used to adjust the rendering of images, backgrounds, and borders

Filter	Description	Example
`blur()`	Adds a Gaussian blur	`filter: blur(5px);`
`brightness()`	Adjusts brightness	`filter: brightness(150%);`
`contrast()`	Increases or decreases contrast	`filter: contrast(120%);`
`grayscale()`	Converts to black & white	`filter: grayscale(100%);`
`invert()`	Inverts colors	`filter: invert(100%);`
`opacity()`	Adjusts transparency	`filter: opacity(70%);`
`saturate()`	Adjusts color saturation	`filter: saturate(200%);`
`sepia()`	Gives an old photo (brownish) effect	`filter: sepia(80%);`
`drop-shadow()`	Adds shadow similar to `box-shadow`	`filter: drop-shadow(10px 10px 5px gray);`

Background-Image:

Property	Possible Values	Description
background-image	`url("image.jpg")`, `none`, `linear-gradient()`, `radial-gradient()`	Sets one or more background images for an element.
background-repeat	`repeat`, `no-repeat`, `repeat-x`, `repeat-y`, `space`, `round`	Defines how the background image repeats horizontally or vertically.
background-position	`left`, `right`, `top`, `bottom`, `center`, or `x% y%`	Sets the starting position of the background image.
background-size	`auto`, `cover`, `contain`, `<length>`, `<percentage>`	Specifies how the background image should be scaled.
background-attachment	`scroll`, `fixed`, `local`	Determines whether the background image scrolls with the page or stays fixed.
background-origin	`border-box`, `padding-box`, `content-box`	Defines where the background image starts painting inside the element box.
background-clip	`border-box`, `padding-box`, `content-box`, `text`	Specifies how far the background should extend (useful with `background-clip: text`).
background-color	Any color value (`red`, `#ff0000`, `rgb()`, `transparent`)	Sets the background color behind the image.
background-blend-mode	`multiply`, `screen`, `overlay`, `darken`, `lighten`, etc.	Blends the background image with the background color.
background	Shorthand property	Allows setting multiple background properties in a single line.

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Background Fixed Example</title>
    <style>
        body {
            margin: 0;
            height: 200vh;
            background-image: url("https://picsum.photos/1200/800");
            background-repeat: no-repeat;
            /* background-position: center; */
            background-size: cover;
            background-attachment: fixed;
        }
    </style>
</head>

<body>
</body>

</html>
```

Buttons:

The `<button>` HTML element is an interactive element activated by a user with a mouse, keyboard, finger, voice command, or other assistive technology. Once activated, it then performs an action, such as submitting a form or opening a dialog.

```html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CSS Buttons</title>
    <style>
        button,
        input[type=button] {
```

```
            border: none;
            padding: 10px 20px;
            color: white;
            font-size: 16px;
            margin: 5px;
            cursor: pointer;
            border-radius: 5px;
        }

        .primary {
            background: #1500ff;
        }

        .danger {
            background: #f44336;
        }

        .info {
            background: #2196F3;
        }

        .warning {
            background: #ff9800;
        }

        .success {
            background-color: #28A745;
        }

        button:hover {
            opacity: 0.8;
        }
    </style>
</head>

<body>

    <button class="primary">Primary</button>
    <button class="danger">Danger</button>
    <!-- <button class="info">Info</button> -->
    <!-- <button class="warning">Warning</button> -->
    <input type="button" value="Info" class="info">
    <input type="button" value="Warning" class="warning">
    <input type="button" value="Success" class="success">
```

```
</body>

</html>
```

User Interface:

User Interface (UI) properties are used to manage how a web page's elements **interact with the user**.
These properties define how users can **click, focus, select, resize, or navigate** elements visually.

CSS User Interface Properties

They are used to:

- Control **user interactivity** (e.g., can the user select text?)
- Change **cursor style**
- Define **focus outlines** (for accessibility)
- Control **appearance** of form controls
- Handle **resize**, **overflow**, and **scrolling**
- Provide **visual feedback** on hover, focus, or disabled states

Property	Function	Example
`cursor`	Sets the style of the mouse pointer	`cursor: pointer;`
`resize`	Lets users resize elements like `<textarea>`	`resize: both;`
`outline`	Draws a border around focused elements	`outline: 2px solid blue;`

user-select	Enables or disables text selection	`user-select: none;`
caret-color	Defines the color of the text cursor in inputs	`caret-color: red;`
appearance	Removes or changes the browser's default element style	`appearance: none;`
pointer-events	Controls whether an element can respond to mouse clicks	`pointer-events: none;`

```
<!DOCTYPE html>
<html lang="en">
<head>
    <style>
        body {
            font-family: Arial, sans-serif;
            padding: 30px;
        }

        button {
            cursor: pointer;
            padding: 10px 20px;
            border: none;
            background: #28a745;
            color: white;
            border-radius: 6px;
        }

        button:disabled {
            opacity: 0.6;
            cursor: not-allowed;
        }
```

```
        textarea {
            resize: both;
            width: 200px;
            height: 100px;
        }

        input {
            caret-color: crimson;
            outline: none;
            border: 2px solid #ccc;
            padding: 6px;
        }

        input:focus {
            border-color: dodgerblue;
        }

        .block {
            user-select: none;
            background: lightgray;
            padding: 10px;
        }
    </style>
</head>

<body>

    <h3>CSS User Interface Properties Example</h3>

    <input type="text" placeholder="Click or focus here">
    <br><br>

    <textarea>Try resizing me</textarea>
    <br><br>

    <button>Enabled Button</button>
    <button disabled>Disabled Button</button>
    <br><br>

    <div class="block">Text selection is disabled here</div>
</body>
</html>
```

Box-Sizing :

The **`box-sizing`** property in CSS defines how the total width and height of an element are calculated — whether they include the element's **padding** and **border** or not.
By default, when you set the width and height of an element, **CSS does not include padding or border** in those measurements — which can cause layout issues.

Syntax : box-sizing: content-box | border-box | inherit;

Value	Description
`content-box` (default)	Width and height include **only the content area**. Padding and border are **added outside** of it.
`border-box`	Width and height include **content + padding + border**. This makes it easier to size elements precisely.
`inherit`	Inherits the `box-sizing` value from its parent element.

Example :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Document</title>
    <style>
        div{
            border : 1px solid;
            width: 500px;
            margin: 5px;
            padding: 4px;
            font-size : 25px;
        }

        .b-box{
            box-sizing: border-box;
        }

        .c-box{
            box-sizing: content-box;
        }
    </style>
```

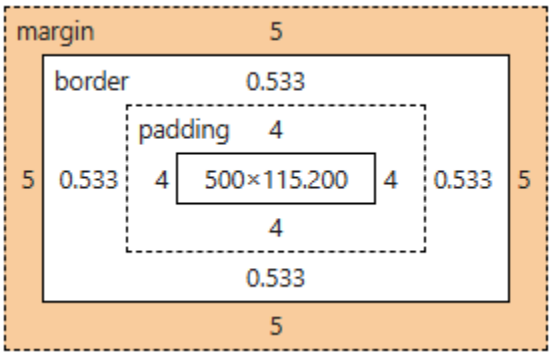
```
</head>
<body>
    <div class="c-box">
        content box = width(500px) + padding(4+4) + border(0.5 +0.5)
        <br>
        sum of all 3 will be greater then 500px
        <br>
        Layout may break if there is not available space
    </div>



    <div class="b-box">
        border box = width( < 500px ) + padding(4+4) + border(0.5 +0.5)
        sum of all 3 will be equal to 500px
        Layout will not break
    </div>

</body>
</html>
```

Box-sizing : content-box



Box-sizing : border-box